Typical protein / antibody screening workflow

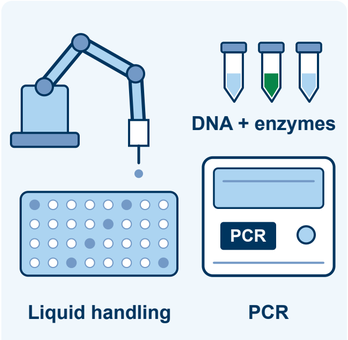
Flat experimental schematics: the tools, inputs and suppliers behind each step

BUILD · Make physical samples

01 / BUILD

Construct design / DNA synthesis

Make DNA constructs for the selected candidates



Sequence-verified DNA

REPRESENTATIVE SUPPLIERS

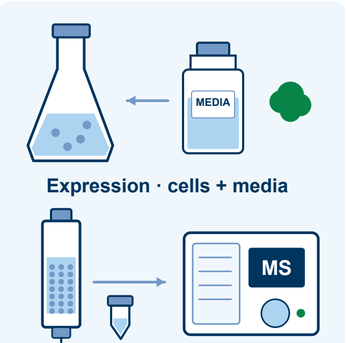
DNA + enzymes
Twist · GenScript · Vazyme

Liquid handling / PCR
Tecan · Hamilton · Thermo Fisher

02 / BUILD

Protein expression & purification

Express the protein, purify it and check sample quality



Purified protein + yield

REPRESENTATIVE SUPPLIERS

Expression / media
GenScript · Gibco

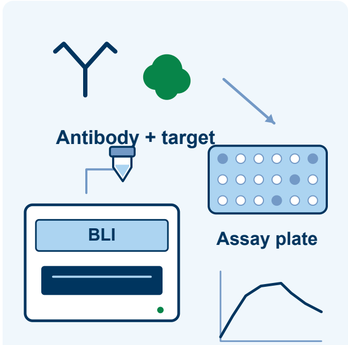
Separation / QC
Cytiva · NanoMicro · Waters

TEST · Generate biological evidence

03 / TEST

Primary binding screen

Identify binders against the chosen target protein



Binding hits

REPRESENTATIVE SUPPLIERS

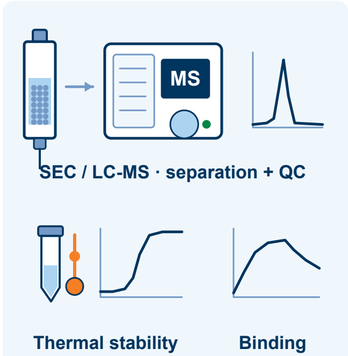
Target reagents
ACROBiosystems · Sino Biological

Binding platforms / labware
Sartorius · Cytiva · JET Biofil

04 / TEST

Multi-property characterization

Measure binding, stability and developability risks



Linked multi-property labels

REPRESENTATIVE SUPPLIERS

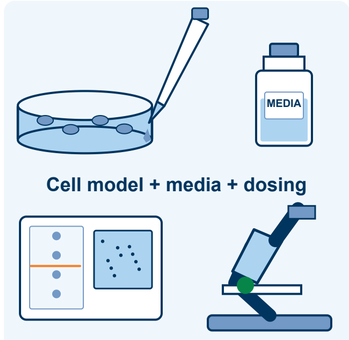
Separation / analytics
Waters · Agilent · NanoMicro

Stability / binding
NanoTemper · Sartorius

05 / TEST

Cell-function validation

Test biological activity in a relevant cell model



Cell potency

REPRESENTATIVE SUPPLIERS

Cell inputs / labware
Gibco · Bio-Techne · Corning

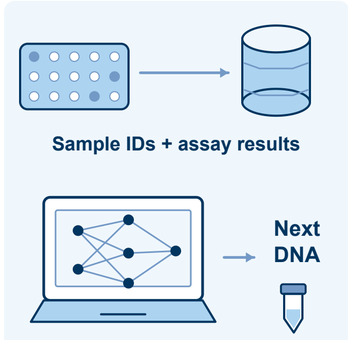
Flow / imaging
Waters Biosciences* · Mol. Devices

LEARN · Iterate

06 / LEARN

Data integration & feedback

Link samples and results to the next model-guided design



Traceable training data

REPRESENTATIVE SUPPLIERS

Automation
Tecan · Hamilton

ELN / LIMS / data
Benchling · Genedata

Linked results → next design → next experiment

Automation + data orchestration span every step

Representative experiments, not a fixed protocol. Assays may run in parallel. LC-MS is shown for analytical QC.

*Waters Biosciences is the former BD Biosciences portfolio. Flat schematics and text are editable; supplier examples and sources: workflow/SOURCES.md.